

^{50}Sn : 50 protons, 76 neutrons (**not** doubly magic)

- 30** **A** From the figure, background count rate can be estimated to be 8 s^{-1} .
Subsequently, half-life can be estimated as the time duration for the count rate to decrease from 58 (=50+8) to 33 (=25+8). Best option is 60 s.